Set 23: Employee Performance Analysis

DSA – Data Handling and Visualization | Lab Experiment

R.No	Reg. No	Name
14	192524098	Divyasree Muthukumar

Aim

To analyse employee performance using a department bar chart, a performance trend line chart and a labelled table in R, and a Tableau dashboard combining a bar chart and a scatter plot.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
library(patchwork)
options(scipen = 999)
# Plain white table theme with black borders (no colours)
tbl_theme <- ttheme_minimal(
  core    = list(fg_params = list(col = "black", fontsize = 11),
                 bg_params = list(fill = "white", col = "black")),
  colhead = list(fg_params = list(col = "black", fontface = "bold", fontsize = 11.5),
                 bg_params = list(fill = "white", col = "black")))
emp <- data.frame(
  Employee_ID       = 1:5,
  Department        = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service  = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)
write.csv(emp, "Employee_Performance.csv", row.names = FALSE)  # for Tableau
dept_pal <- c(HR = "#009E73", Marketing = "#E69F00", Sales = "#0072B2")
```

Q1. Bar Chart – Employees across Departments

> *In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.*

R Program:

```
dept <- as.data.frame(table(emp$Department)); names(dept) <- c("Department", "Employees")
print(dept, row.names = FALSE)
```

```
p_bar <- ggplot(dept, aes(x = Department, y = Employees, fill = Department)) +
  geom_col(width = 0.55) + geom_text(aes(label = Employees), vjust = -0.5, size = 4.5) +
  scale_fill_manual(values = dept_pal) + scale_y_continuous(limits = c(0, 3), breaks = 0:3) +
  labs(title = "Distribution of Employees across Departments", x = "Department", y = "Number
of Employees") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5),
legend.position = "none")
print(p_bar)
ggsave("s23_q1.png", p_bar, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
 Department Employees
         HR         2
  Marketing         1
      Sales         2
```

Output:

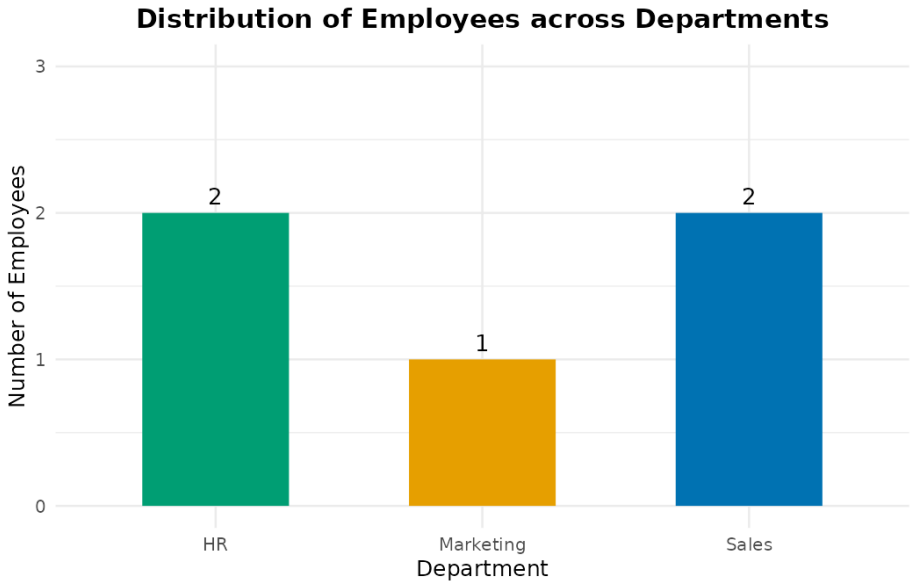


Figure 1: Number of employees per department

Inference:

Sales and HR have 2 employees each (40% each), and Marketing has 1 (20%).

Q2. Line Chart – Performance Trend over Time

> *In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.*

R Program:

```
emp_sorted <- emp[order(emp$Years_of_Service), ]
print(emp_sorted, row.names = FALSE)
cat("Correlation (Years of Service vs Score):", round(cor(emp$Years_of_Service,
emp$Performance_Score), 3), "\n")
print(aggregate(Performance_Score ~ Department, data = emp, FUN = mean))
p_line <- ggplot(emp_sorted, aes(x = Years_of_Service, y = Performance_Score)) +
  geom_line(colour = "grey40", linewidth = 1) +
  geom_point(aes(colour = Department), size = 4.5) +
  geom_text(aes(label = paste0("Emp ", Employee_ID, ": ", Performance_Score)), vjust = -1.3,
size = 3.6) +
  scale_colour_manual(values = dept_pal) +
```

```
  scale_x_continuous(breaks = 1:8, limits = c(1.5, 7.5)) + scale_y_continuous(limits = c(70,
100)) +
  labs(title = "Employee Performance Trend over Years of Service", x = "Years of Service", y =
"Performance Score", colour = "Department") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_line)
ggsave("s23_q2.png", p_line, width = 7.5, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
 Employee_ID Department Years_of_Service Performance_Score
           5         HR                2                76
           2         HR                3                92
           4      Sales                4                90
           1      Sales                5                85
           3  Marketing                7                78
Correlation (Years of Service vs Score): -0.169
  Department Performance_Score
1         HR              84.0
2  Marketing              78.0
3      Sales              87.5
```

Output:

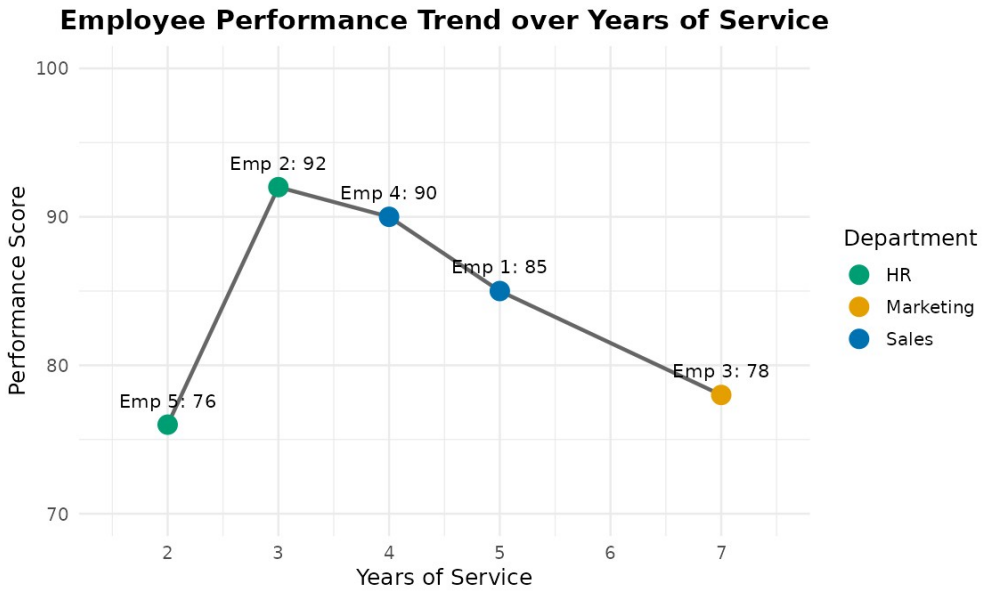


Figure 2: Performance score against years of service

Inference:

The dataset has no dates, so **Years of Service** is used as the time axis. There is no clear trend: scores rise from 76 (2 years) to 92 (3 years), then fall to 78 (7 years). The correlation between service and score is weak and negative (r = −0.17).

By department, Sales has the highest average score (87.5), followed by HR (84.0) and Marketing (78.0). The two HR employees differ a lot (92 and 76), so tenure alone does not explain performance.

Q3. Tableau Dashboard – Bar Chart and Scatter Plot

> *In Tableau, build a dashboard combining a bar chart showing department distribution and a scatter plot displaying the relationship between years of service and performance scores.*

R Program:

```
p_sc <- ggplot(emp, aes(x = Years_of_Service, y = Performance_Score, colour = Department)) +
  geom_point(size = 5) +
  geom_text(aes(label = paste("Emp", Employee_ID)), vjust = -1.4, size = 3.4, colour =
"black") +
  scale_colour_manual(values = dept_pal) +
  scale_x_continuous(limits = c(1, 8)) + scale_y_continuous(limits = c(70, 100)) +
  labs(title = "Years of Service vs Performance", x = "Years of Service", y = "Performance
Score") +
  theme_minimal(base_size = 11) + theme(plot.title = element_text(face = "bold"))
dash <- (p_bar | p_sc) +
  plot_annotation(title = "Employee Performance Dashboard", theme = theme(plot.title =
element_text(face = "bold", size = 15)))
ggsave("s23_dash.png", dash, width = 12, height = 4.8, dpi = 130, bg = "white")
ggsave("s23_q3_scatter.png", p_sc, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Output:

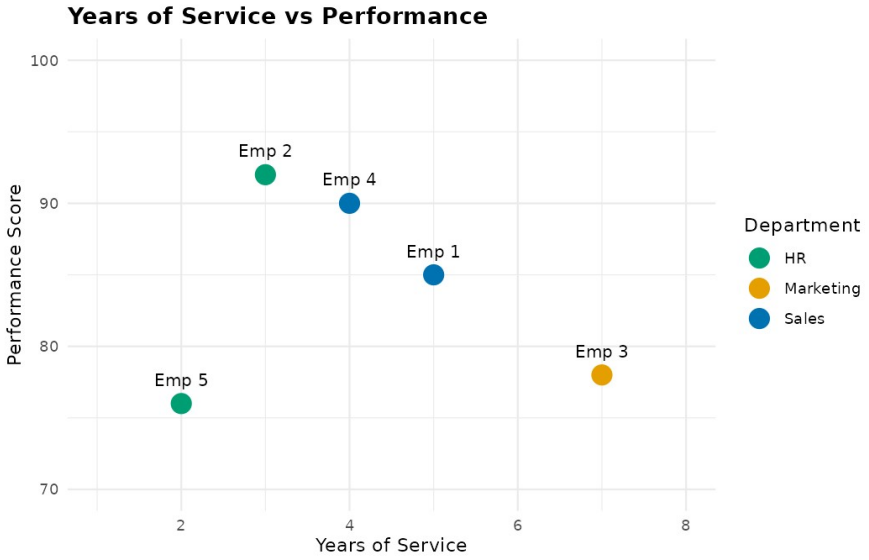


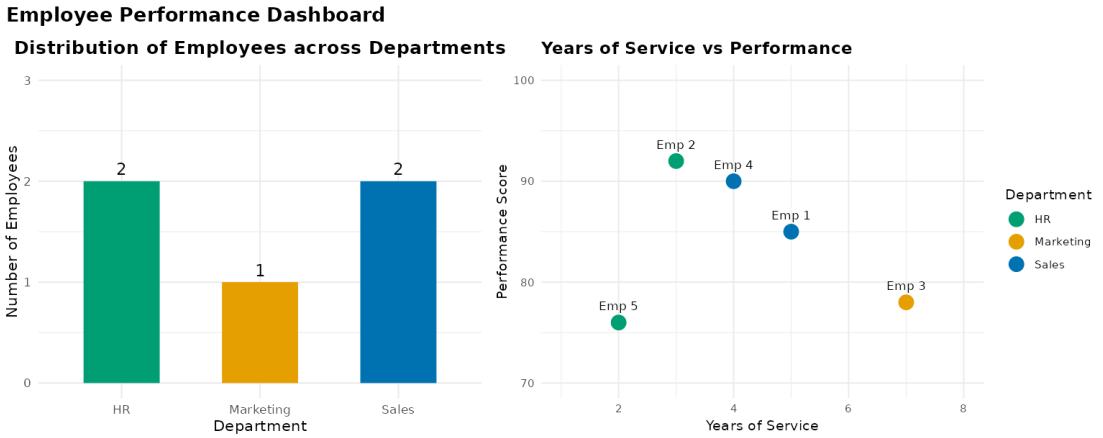
Figure 3: Years of service vs performance score, coloured by department

Procedure in Tableau:

1. Open Tableau → **Connect → Text file** → **Employee_Performance.csv**. Convert **Employee ID** to a Dimension.
2. **Sheet 1 – "Department Distribution":** drag Department to Columns and **CNT(Employee ID)** to Rows. Drag Department to Color and show labels.
3. **Sheet 2 – "Service vs Performance":** drag Years of Service to Columns and Performance Score to Rows. Turn off **Analysis → Aggregate Measures**, or drag Employee ID to **Detail** so each employee is one point. Drag Department to Color and Employee ID to Label. Optionally add a **Trend Line**.
4. From the menu choose **Dashboard → New Dashboard** and set **Size → Automatic** (or Fixed 1200 × 800).
5. Drag the sheets onto the canvas: Department Distribution (left) and Service vs Performance (right). Add a **Text** object at the top as the dashboard title.
6. On each chart, open the sheet's drop-down and select **Use as Filter**, so clicking a bar, point or slice filters the other views.
7. Go to **Dashboard → Actions → Add Action → Highlight** (Run on: Hover) so related marks highlight across all views.
8. Right-click each quick filter → **Apply to Worksheets → All Using This Data Source** (Employee_Performance) so one filter controls every view.

9. Save as a packaged workbook (**File → Save As → .twbx**) or publish to Tableau Public.

Dashboard Layout:



Reference layout of the dashboard (drawn in R). The interactive version is built in Tableau using the procedure above.

Inference:

Clicking the "HR" bar highlights the two HR employees in the scatter plot (scores 92 and 76), so department-level and individual-level views can be compared together.

Q4. Table – Performance Data for Each Employee

> *In R, develop a table showing the performance data for each employee. Label the table elements.*

R Program:

```
tbl <- emp
names(tbl) <- c("Employee ID", "Department", "Years of Service", "Performance Score")
print(tbl, row.names = FALSE)
tg <- tableGrob(tbl, rows = NULL, theme = tbl_theme)
png("s23_q4.png", width = 1000, height = 360, res = 150)
grid.arrange(grid::textGrob("Employee Performance Data", gp = grid::gpar(fontsize = 14,
fontface = "bold")),
             tg, ncol = 1, heights = c(0.25, 1))
invisible(dev.off())
```

Console Output:

```
 Employee ID Department Years of Service Performance Score
           1      Sales                5                 85
           2         HR                3                 92
           3  Marketing                7                 78
           4      Sales                4                 90
           5         HR                2                 76
```

Table Output:

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Inference:

Employee 2 (HR) has the highest score (92) and Employee 5 (HR) the lowest (76). The average score is 84.2.

Result

The employee performance data was visualised with a bar chart, line chart and labelled table in R, and a Tableau dashboard was designed. Sales has the highest average score (87.5), and years of service show only a weak negative relationship with performance (r = −0.17).